

EXHALE: Convergence Reliability and the Steady-State Newton (JFNK) Solver

(work log, 2026-06-10/11)

This memo is the complete record of two tightly linked developments in EXHALE: (i) the discovery that the historical convergence criterion ($du < 10^{-3}$ on the mass-flux spread) can stop runs at premature, non-steady snapshots — the WASP-121b reference used throughout earlier work was such a snapshot, low by ~ 0.08 dex in $\dot{M}$ — together with the new convergence diagnostics that fix this; and (ii) the construction, debugging, and first converged run of a direct steady-state solver (pseudo-transient continuation + Jacobian-free Newton–Krylov), which reaches residual levels unattainable by time marching at $\sim 1/200$ of the cost. Every code change is opt-in: with all new options at their defaults, the marching code is byte-identical to the pre-existing behavior (verified by the regression harness after every increment). Statements of physical interpretation below are tentative unless explicitly verified.

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1. Background and Stage-I refactor (context)

The work below sits on top of the Stage-I “extensibility” refactor of the same dates (named species constants; the composition single policy point for n_e/n_{tot} /EOS; schema-2 output headers and a label-mapping `load_IC` that preserves

the metal ionization state on restart; element-loop abundances; rate dispatch by ion index). Each step was a no-physics-change refactor gated by a deterministic regression harness (regression/run_check.sh: single-threaded WASP-121b runs, golden outputs compared byte-wise excluding # header lines; matrix cases wasp_full = He 2³S + metals and wasp_he23off). Details: docs/refactor_stage1_log.md. The same harness gates every solver-related change in this memo.

2. The convergence-criterion problem

2.1 What du measures, and what it misses

ATES declares steady state when

$$du = \frac{\max_j |\rho v r^2|_j - \min_j |\rho v r^2|_j}{\min_j |\rho v r^2|_j} \Big|_{j \in [j_{\min}, N]} < du_{\text{th}}, \quad du_{\text{th}} = 10^{-3}, \quad (1)$$

i.e. the *spatial spread* of the mass flux over the wind window ($r \geq r_{\text{esc}}$). Two blind spots were demonstrated this session:

1. du cannot see a *uniform level drift*: the whole $\rho v r^2$ profile can rise or fall while staying flat.
2. du is *oscillatory* on WASP-121b: it dips below 10^{-3} transiently long before the wind settles.

2.2 The premature-dip discovery (WASP-121b)

The single-thread regression golden (He 2³S + metals, two-stage PLM→WENO3, stopped by $du < 10^{-3}$ at step 7,288) has raw $\log_{10} 4\pi \rho v r^2(r_{\text{max}}) = 13.631$. Suppressing the du stop and marching on, the flux level rises *monotonically* (the windowed level change lev_rel falls $0.23 \rightarrow 2.7 \times 10^{-3}$) toward $\log_{10} \dot{M} \simeq 13.71$: at step 35,085 the state reaches 13.707 and is still creeping upward, with T_{max} falling from 11,021 K to 10,767 K. Three independent paths agree on the deeper state (long cold marching →13.707; warm restarts →13.716; the naive local- dt experiment →13.716 in 1,702 steps). **The old golden was a premature du -dip snapshot, not the steady state** — approximately 0.08 dex ($\sim 20\%$) low in $\dot{M}$.

Two earlier conclusions were corrected as a result. First, naive per-cell pseudo-time stepping (“Time stepping: Local”) had been judged “answer-shifting”; in fact it reaches the *correct deeper* state $\sim 20\times$ faster — the judgment was an artifact of comparing against the premature golden. (The LTS state does carry larger per-cell hydro residuals, §??, so it is “fast but needs polishing.”) Second, the apparent “path-dependent quasi-steady states” largely collapse to one true state plus premature snapshots along the way.

2.3 New diagnostics and stops (all opt-in)

Level-stability gate (Level1 tol:). A converged/stalled stop additionally requires the mean $|\rho v r^2|$ over $[j_{\min}, N]$ to change by less than lev_th (relative) across the last N_{stall} steps (ring buffer in EXHALE_main; the relative level change is now the 4th column of the per-step stdout). This correctly refuses the premature dip (the cold run marches past step 7,288), but a practical tolerance cannot discriminate ultra-slow drifts; it is necessary, not sufficient. Default off.

Steady-residual diagnostic (EXHALE_RESIDUAL=1). The finite-volume steady residual

$$\mathbf{R}_j = \begin{cases} (\nabla \cdot \mathbf{F})_j - S_j, & \text{mass, momentum,} \\ (\nabla \cdot \mathbf{F}_E)_j - S_{E,j} - (\mathcal{H}_j - \mathcal{C}_j), & \text{energy,} \end{cases} \quad (2)$$

evaluated once on a loaded state (one Reconstruct+RK_rhs pass; heating/cooling from ioniz_eq; reported as $\max_j |\mathbf{R}| / \max_j |u|$ per component over the window). This is *independent* of du/dt /stall and ranks candidate states:

state	R_{mass}	R_{mom}	R_{energy}
golden 13.631 ($du < 10^{-3}$ stop)	9.9×10^{-2}	4.7×10^{-1}	3.1×10^1
warm 13.716 (local-dt + cont.)	2.4	2.8	7.4×10^{-1}
cold 13.707 (35k steps)	2.1×10^{-2}	1.6×10^{-2}	1.5×10^{-2}

The premature golden has an energy residual of ~ 30 : heating and cooling are grossly out of balance, which du cannot detect. (The energy value 1.5×10^{-2} for the cold state corresponds to the consistent ghost-handling order, §3.1.)

Residual-based stop (Resid tol:). In stage 2 the run converges on $\max_k \max_j |\mathbf{R}_{j,k}| / \max_j |u_k| < \text{resid_th}$ (evaluated every $N_{\text{resid}}=500$ steps; cheap), *replacing* the du test. Validation: a run with $\text{resid_th} = 10^{-2}$ reached $du = 8.6 \times 10^{-4}$ (below the old stop) at step 32k and correctly kept marching because R_{max} was still 3.1×10^{-2} and falling monotonically. Default off.

3. The steady-state solver

The marching cost to a genuinely converged wind ($> 35\text{k}$ steps for WASP-121b; $> 10^6$ steps for HD189733b without reaching it) motivates solving the steady residual directly: $F(Y) = 0$.

3.1 Formulation

Unknowns $Y \in \mathbb{R}^{3N}$: the conserved hydro variables $(\rho, \rho v, E)$ on the N physical cells. Ghost cells are not unknowns; Apply_BC fills them from the interior on every residual evaluation (lower boundary: pinned ρ_{bc} , $p = 1 + dp_{\text{bc}}$ and the one-way velocity valve; upper: extrapolation). **Local elimination:** given Y , the ionization equilibrium is re-solved (ioniz_eq) and the resulting heating/cooling enter the energy residual — so F is a pure-hydro residual of size $3N$ (“method A” of the SUNDIALS plan), and the non-local radiation (column densities) is inside F . The residual construction is shared with the diagnostics (steady_residual.f90: assemble_residual); the vector form lives in steady_newton.f90 (pack_U/unpack_U/eval_residual), validated to reproduce the EXHALE_RESIDUAL numbers to all digits. One ordering subtlety: Apply_BC must run *before* the chemistry/heating step so that F depends only on the interior Y ; the diagnostic hook was aligned to the same order.

3.2 Jacobian and linear algebra

Banded approximation. With WENO3’s ± 2 -cell stencil and 3 coupled variables per cell, $\partial F_i / \partial Y_j$ has half-bandwidths $k_l = k_u = 8$ in the flat ordering. A graph coloring with stride $k_l + k_u + 1 = 17$ recovers all band entries of a *local* residual

from 17 finite-difference probes. The banded matrix is factored with LAPACK dgbtrf/dgbtrs (LDLIBS=llapack; ATLAS LAPACK 3.7.1 on this machine, link-tested). A unit test (EXHALE_JAC_TEST=1) checks Jr against a directional finite difference: relative mismatch 1.3×10^{-4} , i.e. the band layout is correct (an error would be $O(1)$).

JFNK. Because the radiation makes F weakly non-local, the banded matrix is used as a *preconditioner* while the true Jacobian-vector product is matrix-free, $Jv \simeq [F(Y + \varepsilon v) - F(Y)]/\varepsilon$ (jv_product), inside a right-preconditioned GMRES(m) (pgmres, modified Gram–Schmidt + Givens; inexact-Newton forcing 10^{-1}). The outer loop is pseudo-transient continuation,

$$\left(\frac{1}{\Delta\tau}I + J\right)\delta Y = -F(Y), \quad \Delta\tau \leftarrow \Delta\tau \max(\lambda, 0.1) \frac{\|F\|_{\text{old}}}{\|F\|_{\text{new}}} \quad (\text{SER ramp}), \quad (3)$$

with a backtracking line search and positivity guards ($\rho, p > 0$).

3.3 The five necessary ingredients (each found by a stall)

The path to convergence was a sequence of well-localized stalls; each fix was verified by its own probe (warm start from the cold-35k state, $\Delta\tau_0 = 1$ unless noted). Removing any one ingredient reproduces the corresponding stall.

1. **Smooth line-search merit.** The first driver used the component-wise max-relative residual as the merit; it is non-smooth and rejected every step ($\lambda \rightarrow 10^{-6}$). Fix: merit = $\|F\|_2$ (later $\|D^{-1}F\|_2$). After the fix the banded-PTC driver descends, but only linearly ($\sim 2\%$ /iteration) — and a $\Delta\tau_0 = 1$ probe showed a textbook Newton step (residual halved in one step) before stalling, proving the machinery and isolating the next blocker.
2. **Matrix-free Jv (JFNK).** The banded Jacobian omits the non-local radiation response; with exact Jv via GMRES the merit fell $4.5\times$ in two steps (vs. 2% /step), $\Delta\tau$ grew. Stall again — now localized (whole-domain worst-cell diagnostic) to the *base* cells $j = 1-2$ (an earlier “sonic point at r_{esc} ” reading was an artifact of searching only $[j_{\min}, N]$; the sonic point is actually at $r = 1.61$ and r_{esc} enters no physics). Freezing base cells with anchor rows does *not* help: the worst cell simply follows the slab edge and the anchor interface itself introduces a kink.
3. **Diagonal scaling.** The unknowns span ~ 7 decades (base $\rho \sim 1$ vs. wind $\sim 10^{-7}$ in code units); FD steps with absolute floors give wind cells 15–60% relative perturbations and corrupt exactly the columns where the residual lives after the first base-scale corrections. With $D_i = \max(|Y_i|, 10^{-6} \max_j |Y^{(k)}|)$, the scaled system $D^{-1}(\frac{1}{\Delta\tau}I + J)D$, D-relative FD steps, and merit $\|D^{-1}F\|_2$: GMRES needs 2–6 vectors (was 9) and the merit falls $24\times$ before stalling — at the base again.
4. **Smooth base valve.** The lower BC ghost velocity is $\max(v_1, 0)$, and the steady state hugs the kink ($v_1 \simeq +4.5 \times 10^{-3} c_s$): Newton trial steps crossing $v_1 = 0$ are non-differentiable. Opt-in replacement (Valve eps:): $v_g = \frac{1}{2}(v + \sqrt{v^2 + \varepsilon_v^2})$, smooth, $\rightarrow 0$ for $v \ll -\varepsilon_v$, $\rightarrow v$ for $v \gg \varepsilon_v$ (an earlier variant with a $-\varepsilon_v/2$ shift leaked inflow and was fixed). With $\varepsilon_v = 10^{-2}$ (too large, it perturbs the state) the solver descends $10\times$ *through* the base — differentiability is what Newton needed — but $\varepsilon_v = 10^{-4}$ still stalled at the base: the valve was not the only kink.
5. **Frozen WENO weights + non-monotone line search.** The ESWENO3 smoothness factors $S_{0,1} = 1 + \tau^2/\beta_{0,1}$ are smooth but strongly nonlinear; freezing them at the outer iterate (weno_mode: store on a mode-1 pass, reuse

on mode-2 passes for all Jacobian probes, Jv products, and line-search trials — the standard lagged-weights remedy) removes their response from the inner problem. Independently, the monotone Armijo test fails once the required decrease $10^{-4}\lambda\|F\|$ falls below the iterative-chemistry noise floor of the residual; a *non-monotone* (Grippo, memory 5) acceptance — improve on the worst of the last five accepted merits — is the standard remedy.

3.4 Result: first converged steady solve

With all five ingredients (warm start from the cold-35k state, $\Delta\tau_0 = 1$, $\varepsilon_v = 10^{-4}$):

target	iterations	final $\ R\ $
Resid tol = 10^{-3} 8 (all $\lambda = 1$, GMRES 2–3)		9.5×10^{-4}
Resid tol = 10^{-4}	22	6.0×10^{-5}

Each outer iteration costs ~ 25 residual evaluations (17 for the banded preconditioner + GMRES + line search), so the 10^{-3} solve costs the equivalent of ~ 200 marching steps — versus $> 35,000$ marching steps that never reached this residual level. Convergence is linear, not quadratic (the frozen weights are refreshed per outer iteration — a Picard–Newton hybrid — and the chemistry noise floor remains); every step is a full Newton step.

Physical validation (vs. the marching references):

state	$\log_{10} \dot{M}_{\text{raw}}$	spread($r \geq 1.5$)	T_{max}
Newton (8 its)	13.7111	5.0×10^{-4}	10,747 K
cold marching (35k steps)	13.7066 (rising)	1.2×10^{-3}	10,767 K
old golden (premature)	13.6309	1.8×10^{-3}	11,021 K

The Newton solution sits on the marching asymptote ($\simeq 13.71$), is the flattest state produced to date, and agrees with the deep-marching reference to 0.005 dex in $\dot{M}$ and 0.2% in T_{max} .

4. Runtime options and hooks (reference)

All defaults preserve the legacy behavior byte-identically (re-verified by the harness after every change in this memo).

input line	default	effect
du_th [PLM,WENO3]: a b	—	convergence thresholds; the single-stage vs. two-stage choice is set by Reconstruction scheme: (PLM/WENO3 single-stage use only a ; PLM+WENO3 two-stage: PLM until $du < a$ or stall, then WENO3 until $du < b$)
Stall [tol,N]: t n	$10^{-6}, 2000$	stall-detector override
Level tol: v	off	mass-flux level-stability gate (§2.3)
Resid tol: v	off	residual-based convergence (§2.3)
Time stepping: Local	off	per-cell pseudo- dt (steady-state acceleration; reaches the deep state fast but leaves per-cell residuals)
Energy solver: Explicit	semi-implicit	original forward-Euler source update (isolation tests)
Valve eps: v	off (hard $\max(v, 0)$)	smooth base valve (§3.3)
CFL: v	0.6	CFL override

environment hook	effect
EXHALE_DUMP_IC=1	write the loaded state and stop (restart round-trip test)
EXHALE_RESIDUAL=1	evaluate and report the steady residual of the loaded state, then stop
EXHALE_NEWTON_TEST=1	pack/unpack + vector-residual self-test
EXHALE_JAC_TEST=1	banded-Jacobian vs. directional-FD unit test
EXHALE_PTC=1	run the steady solver from the (loaded) IC, write output, stop; EXHALE_PTC_JFNK=1 selects JFNK (else banded PTC); EXHALE_PTC_DTAU0, EXHALE_PTC_NFIX optional

Working recipe for the converged WASP-121b solve:

```
# input.inp: ... + "Valve eps: 1.0e-4" + "Resid tol: 1.0e-3"
# output/{Hydro_ioniz,Ion_species}_IC.txt: a warm-start state
EXHALE_PTC=1 EXHALE_PTC_JFNK=1 EXHALE_PTC_DTAU0=1.0 ./EXHALE.x
```

5. Hardening results (2026-06-11, after first convergence)

HD189733b — 33 s vs. >3 h. Warm-started from the deepest marching state available (1.16M steps, $du = 6.6 \times 10^{-3}$; starting residual $(2.4, 5.2, 2.4) \times 10^{-3}$), the JFNK solver converged to $\|R\| = 9.5 \times 10^{-4}$ in 129 iterations / **33 seconds wall time**. The converged state has $\log_{10} \dot{M}_{\text{raw}} = 10.298$ vs. 10.218 (marching) and 10.223 (pristine ATES-Code-main) — both still drifting upward — and is the flattest state produced (spread 7.6×10^{-3} vs. 1.25×10^{-2}). Tentatively, the true HD189733b $\dot{M}$ is $\sim 20\%$ above the marching estimates, the same premature-snapshot pattern quantified for WASP-121b. Tightening to 10^{-4} grinds: the frozen-weight refresh jumps are large at the breathing base and progress floors at $\|R\| \sim 3.4 \times 10^{-4}$ — the practical floor for this planet with the current settings.

Valve- ε insensitivity. Two full Newton solves of WASP-121b (tol 10^{-4}) with $\varepsilon_v = 10^{-4}$ and 10^{-5} give $\log_{10} \dot{M} = 13.7110$ and $T_{\text{max}} = 10750.2\text{ K}$ *identically* to the digits quoted: the valve smoothing is a pure numerical regularization.

Solver: Newton (production wiring) and the switch criterion. The warm-up→Newton hand-off was wired into the main loop. Two failure modes were found and fixed along the way, both reiterating the central lesson: (i) a du -based switch fails — at $du < 10^{-2}$ during the early transient the residual is still $\|R\| \approx 4.8$, far outside Newton’s basin; (ii) with the switch fixed, the marching’s own $du < 10^{-3}$ stop fired first, returning the premature state. The final design: *the switch and the warm-up termination are both residual-based* (periodic monitor; switch at $\max \|R\| < 5 \times 10^{-2}$, du -stops suspended in this mode). End-to-end validation from a *cold isothermal IC* (WASP-121b, single input line Solver: Newton): two-stage warm-up to step 27,000 ($\|R\| = 4.85 \times 10^{-2}$) → JFNK 16 iterations → $\|R\| = 5.4 \times 10^{-4}$ → standard outputs and post-processing; final $\log_{10} \dot{M}_{\text{raw}} = 13.7111$, spread 5.1×10^{-4} , $T_{\text{max}} = 10,748$ K — identical to the warm-started reference. Wall time 16 min (4 threads), dominated by the warm-up.

Transmission spectra of the Newton solutions. EXHALE_transit.py was run on both converged solutions (after PP-only runs generated the *_adv profiles): He 10830 / Ly α / H α / H β spectra for both planets plus the metal lines for WASP-121b (Mg II 2796: 3.7%, Ca II K: 3.8%, Na I D2: 1.7% line-center). Peak line-center absorptions: WASP-121b He 3.7%, H α 3.7%; HD189733b He 11.9%, Ly α 46%, H α 1.8% (consistent in magnitude with the Jensen et al. 2012 H α detection). Figures and numbers are collected in regression/steady_solver_results.ipynb; usage is documented in the updated user manual (§ TPM).

Failure containment (a genuine JFNK failure, and two fixes). Re-running the manual’s four figure runs through the production pipeline exposed a real failure mode on the spherical, metals-off tutorial planet (a generic hot Jupiter, domain to $10R_p$). The solve nearly converged ($\|R\| = 6.0 \times 10^{-3}$ at iteration 2), but the non-monotone reference still held the huge *initial* merit ($\|F_s\|_2 \simeq 10^4$), so a blow-up step was “accepted”; the search then stalled at $\lambda \simeq 10^{-6}$, accepting zero-length steps against the now-flat reference for ~ 470 iterations, and — worst of all — the main loop accepted the final unconverged state ($\|R\| = 0.42$) as converged and wrote it out. Two fixes (both in the default-off Newton path; legacy byte-identity re-verified): (i) *best-iterate tracking and fail-fast* in solve_steady_jfnk — the best residual state is stored, and the solve aborts with info=2 after 15 outer iterations without a new best, returning the best iterate; (ii) in EXHALE_main, info $\neq 0$ is treated as a failure: the run prints JFNK failed, keeps the best iterate, disables the Newton mode, and *resumes plain time-marching* with the du -based stops re-armed, so an unconverged Newton state can never masquerade as the final answer. Re-run with the fixes, the same configuration aborts after 15 stuck iterations, returns the best iterate ($\|R\| = 5.97 \times 10^{-3}$), and the resumed marching stops one step later ($du = 3.4 \times 10^{-4}$, already below threshold); the run ends on an honest, near-converged state instead of a diverged one. The three other figure runs converged cleanly through the pipeline (tutorial metals-on $\|R\| = 5.6 \times 10^{-4}$, WASP-121b spherical Case A 4.1×10^{-4} , Roche Case B 5.8×10^{-4} , all info=0).

Can JFNK start cold? (basin-boundary experiments, 2026-06-11). Three probes on WASP-121b (all with the fail-fast guard, so each failure is cheap and the run still finishes honestly via the marching fallback):

start of JFNK	start $\ R\ $	JFNK outcome	final answer
cold isothermal IC	1.4×10^4	1 iter $\rightarrow$ 11, hard stall at the base	— (abort)
step 6,500 ($R_{\text{switch}}=0.5$)	0.43	$\rightarrow$ 0.245, stall	fallback <i>du</i> -stop, $\dot{M}$ 13.34 (premature)
step 12,500 ($R_{\text{switch}}=0.1$)	0.091	first step <i>worsens</i> (0.148), stall	fallback <i>du</i> -stop, $\dot{M}$ 13.38 (premature)
step 27,000 (default 5×10^{-2})	0.0485	16 iters $\rightarrow 5.4 \times 10^{-4}$	$\dot{M}$ 13.7111 ✓

The Newton basin boundary on this planet therefore lies between $\|R\| \simeq 0.09$ and 0.049 — the default switch is calibrated essentially at the edge, and raising it buys nothing: every stall pins to the first base cells ($j = 1\text{--}2$, momentum), where the valve/ionization coupling makes the frozen-weight local model invalid far from the fixed point. PTC’s small- $\Delta\tau$ limit is just implicit time stepping at $\sim 20\text{--}60\times$ the cost per unit pseudo-time of an explicit RK step, so a cold-start PTC could not beat the marching warm-up economically even where it survives. Practical caveat: when a Solver: Newton run prints JFNK failed, the fallback finishes on the *du* stop, which can be a premature state (13.34/13.38 vs. 13.71 here) — re-check such runs with EXHALE_RESIDUAL=1. Probes preserved in regression/{jfnk_cold, newton_rsw05, newton_rsw01}.

6. Remaining open items

1. **HD189733b residual floor.** Progress floors at $\|R\| \sim 3.4 \times 10^{-4}$ (frozen-weight refresh jumps at the breathing base); an under-relaxed weight refresh between outer iterations is the natural next step.
2. **Non-monotone reference poisoning.** The tutorial-planet failure above was *contained*, not prevented: the 5-entry Grippo memory can retain a huge early merit and thereby accept a blow-up step. Capping the reference at a multiple of the current merit (or flushing the history after a large drop) should prevent it outright.
3. **Chemistry noise floor.** The non-monotone search works around the `ioniz_eq` iteration noise; tightening the inner solve (or smoothing its switch points) would restore certifiable descent and possibly quadratic tails.
4. **Quasi-steady physics.** The deep states still show slow residual decay rather than an exact fixed point (base breathing); whether an exact steady state of the discrete scheme exists with the hard valve remains open — the smooth-valve fixed point found here is the natural regularized answer.

7. File inventory

file	content
src/modules/time_step/steady_residual.f90	shared steady-residual assembly + norms (diagnostics, monitor, Newton)
src/modules/time_step/steady_newton.f90	pack/unpack, residuals, scaling, banded Jacobians (frozen/full), jv_product, pgmres, solve_steady_ptc, solve_steady_jfnk, base-anchor option
src/modules/states/Reconstruction.f90	weno_mode frozen-weights support (S0/S1 store/reuse)
src/modules/states/Apply_BC.f90	smooth base valve (valve_eps)
src/modules/init/parameters.f90	new option variables (defaults off)
src/modules/files_IO/input_read.f90	option parsing (keyword lines after Force start:)
EXHALE_main.f90	level/residual monitors and stops; env hooks
regression/	harness, goldens, and all probe run directories (ptc_warm, crit_cold, resid_*, ...)
docs/steady_solver_design.md	incremental design notes (§§4–11) with the raw probe logs
docs/refactor_stage1_log.md	Stage-I extensibility change log